

SAINTS Test Fixture - Tutorial 5

Professor Model Solutions and Suggested Rubrics

Use this file as the lecturer's reference. The rubric is intentionally simple so you can test AI-proposed scores, instructor overrides, feedback editing, review status, and final publication.

Q1 - Model Solution [4 marks]

1. $x + 7 = 19$

2. $x = 19 - 7$

3. $x = 12$

Rubric criterion	Marks
Identifies the correct inverse operation	1
Rearranges correctly	1
Obtains $x = 12$	2

Q2 - Model Solution [6 marks]

1. $x^2 - 5x + 6 = 0$

2. $(x - 2)(x - 3) = 0$

3. $x = 2$ or $x = 3$

Rubric criterion	Marks
Chooses a valid solution method	1
Factorises correctly	2
Applies zero-product property	1
Obtains both roots $x = 2$ and $x = 3$	2

Q3 - Model Solution [6 marks]

1. $2x^2 - 8x = 0$
2. $2x(x - 4) = 0$
3. $x = 0$ or $x = 4$

Rubric criterion	Marks
Takes out the common factor correctly	2
Retains both factors / does not divide away x	2
Obtains both roots $x = 0$ and $x = 4$	2

Q4 - Model Solution [4 marks]

1. $x^2 = 49$
2. $x = \pm\sqrt{49}$
3. $x = 7$ or $x = -7$

Rubric criterion	Marks
Recognises both square roots are required	2
Obtains $x = 7$	1
Obtains $x = -7$	1

Q5 - Model Solution [6 marks]

1. $x^2 + x - 12 = 0$

2. $(x + 4)(x - 3) = 0$

3. $x = -4$ or $x = 3$

Rubric criterion	Marks
Chooses a valid solution method	1
Factorises correctly	2
Applies zero-product property	1
Obtains both roots $x = -4$ and $x = 3$	2

Expected testing note: The sample student is correct on Q1 and Q2, misses one root on Q3 and Q4, and uses an incorrect factorisation on Q5. This gives the instructor meaningful marks and feedback to review.